Path With Minimum Effort (medium)

Problem Statement

Imagine you're on a grid of size n x n, where each point has an elevation. Your task is to find a path from the top-left corner to the bottom-right corner that minimizes the effort required to travel between consecutive points, where effort is defined as the absolute difference in elevations between two points. Your goal is to determine the smallest maximum effort required for any path from the first point to the last.

Example Generation

Example 1:

- Input: `[[1,2,3],[3,8,4],[5,3,5]]`
- Expected Output: `1`
- Justification: The path with the minimum effort is along the edges of the grid (right, right, down, down) which requires an effort of 1 between each pair of points.

Example 2:

- Input: `[[1,2,2],[3,3,3],[5,3,1]]`
- Expected Output: `2`
- Justification: The path that minimizes the maximum effort goes through (1,2,2,3,1), which has a maximum effort of 2 (from 3 to 1).

Example 3:

- Input: `[[1,1,1],[1,1,1],[1,1,1]]`
- Expected Output: `0`
- Justification: The path that minimizes the maximum effort goes through (1,1,1,1,1), which has a maximum effort of 0.

Try it yourself

Try solving this question here:

Java

```java
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;

// Edge Class to store the difference between two points
class Edge {
    int x;
    int y;
    int difference;

    Edge(int x, int y, int difference) {
        this.x = x;
        this.y = y;
        this.difference = difference;
    }
}

// Solution Class
public class Solution {
    public int minimumEffortPath(int[][] heights) {
        // ToDo: Write Your Code Here.
```

Test Case | Run Code Result | Stdout

```
[[1,2,2],[3,8,2],[5,3,5]]
```

Use Example Testcases | Run | Submit

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Solution: Path With Minimum Effort

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